and compact. Changing mix design parameters, aggregate source, and/or gradation can improve workability.

Harsh mixtures (mixtures containing a high percentage of coarse aggregate) have a tendency to segregate during handling and may be difficult to compact. To make a mixture more workable, the aggregate gradation can be adjusted by including additional fine aggregate, by changing the source of the fine aggregate to include less crushed fines or by increasing the asphalt content of the mix. Care should be taken to ensure that the rutting resistance of the mix is not compromised in order to provide a workable mix. Many high-strength mixtures are harsh and difficult to compact. Too high a filler content can also affect workability. It can cause the mix to become gummy, making it difficult to compact.

Workability is especially important when hand placement and raking (luting) around manhole covers, sharp curves, and other obstacles are required. It is important that mixtures used in such areas be highly workable.

Mixtures that can be too easily worked or shoved are referred to as tender mixes. Tender mixes are too unstable to place and compact properly. They are often caused by:

- A shortage of mineral filler
- Too much medium-size sand
- Smooth, rounded aggregate particles
- Too much moisture in the mix

Although not normally a major contributor to workability problems, asphalt does have some effect on workability. Because the temperature of the mix affects the viscosity of the asphalt, a temperature that is too low will make a mix unworkable and a temperature that is too high may make it tender. Asphalt grade may also affect workability, as may the percentage of asphalt in the mix.

4.1.3 Hot Mix Asphalt Mixture Behavior

When a sample paving mixture is prepared in the laboratory, it can be analyzed to determine its probable performance in a pavement structure. The analysis focuses on the following five characteristics of the mixture and the influence those characteristics are likely to have on mix behavior:

- Mix density
- Air-voids
- Voids in the mineral aggregate
- Voids filled with asphalt
- Asphalt content

Mix Density

The density of the compacted mix is its unit weight or the weight of a specific volume of mix. Low unit weights are often associated with poor quality aggregates that may lead to pavement performance problems.

In mix design testing and analysis, density of the compacted specimen is usually expressed in pounds per cubic foot (lb/ft^3; kilograms per cubic meter [kg/m^3]). It is calculated by multiplying the bulk specific gravity of the mix by the density of water (62.4 lb/ft^3; 1 g/cm^3). The bulk density and the maximum theoretical density, both of which are determined in the laboratory, are used as standards to determine field compaction levels.

Air-Voids

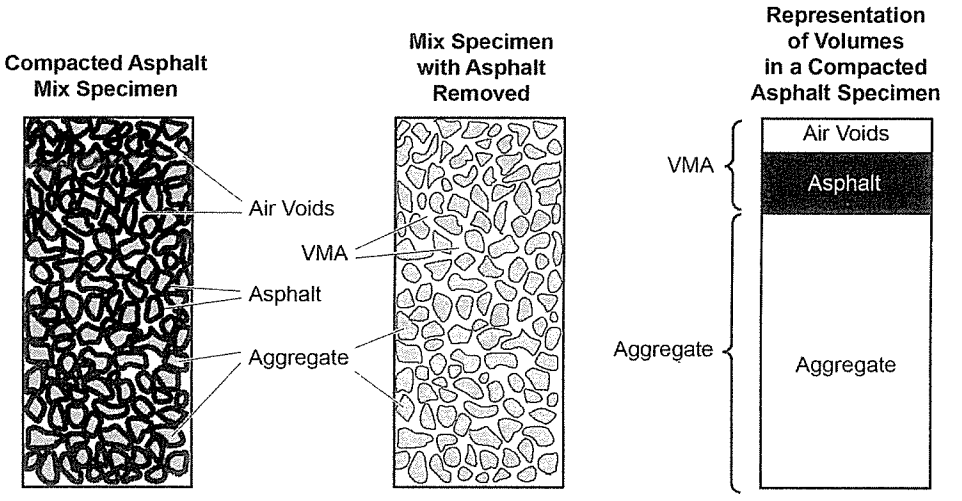
Air-voids (Figure 4.2) are small pockets of air between the coated aggregate particles in the final compacted HMA. A certain percentage of air-voids is necessary in the finished HMA to allow for a slight amount of compaction under traffic and a slight amount of asphalt expansion because of temperature increases. In general, the air-void level used for the design of HMA is 4 percent. For special uses, a lower design air-void content may be specified. Job specifications usually require that pavement compaction achieve an in-place air-void content of less than 8 percent and more than 3 percent.

The durability of an asphalt pavement is also a function of the air-void content of the in-place HMA pavement. The lower the in-place air-voids, the less permeable the mixture becomes. An air-void content that is too high provides passageways through the mix that allow damaging air and water to enter. An air-void content that is too low can lead to flushing or bleeding.

Density and in-place air-void content are directly related. The higher the density, the lower the voids in the mix will be, and vice versa.

Voids in the Mineral Aggregate

Voids in the mineral aggregate (VMA) are the intergranular void spaces that exist between the aggregate particles in a compacted paving mixture. VMA includes air-voids and spaces filled with asphalt. VMA (Figure 4.2) is



Note: For simplification, the volume of absorbed asphalt is not shown

Figure 4.2 Voids in a Compacted HMA Mixture

a volumetric measurement expressed as a percentage of the total volume of a compacted mix.

VMA represents the space that is available to accommodate the effective volume of asphalt (i.e., all of the asphalt that is not absorbed into the aggregate) and the volume of air-voids necessary in the mixture. The more VMA in the aggregate, the more space is available for the films of asphalt. The durability of the mix increases with the film thickness on the aggregate particles. Therefore, specific minimum requirements for VMA are recommended and are specified as a function of the aggregate size.

Minimum VMA is necessary to achieve an adequate asphalt film thickness, which results in a durable asphalt pavement. HMA mixtures with below-minimum VMA values will have thin films of asphalt and will provide an HMA pavement with low durability. Therefore, economizing in asphalt content by lowering VMA is actually counterproductive and detrimental to pavement quality.

Voids Filled with Asphalt

Voids filled with asphalt (VFA) is the percentage of intergranular void space between the aggregate particles (VMA) that contain or are filled with asphalt.

VFA is used to ensure proper asphalt film thickness in the mix. If it is too low the mix will have poor durability or if it is too high the mix can be unstable. The acceptable range of VFA varies depending upon the traffic level for the facility. Higher traffic requires a lower VFA, because mixture strength and stability is a greater concern. Lower traffic requires a higher range of VFA to increase HMA durability. A VFA that is too high, however, may yield a plastic mix.

Asphalt Content

The asphalt content in the mixture is critical. It must be accurately determined in the laboratory, and then precisely controlled on the job. The asphalt content for a particular mix is established by using the criteria specific to the mix design method.

The optimum asphalt content of a mix is highly dependent on aggregate characteristics such as gradation and absorption. Aggregate gradation is directly related to optimum asphalt content. The finer the mix gradation, the larger the total surface area of the aggregate will be and a greater amount of asphalt will be required to uniformly coat the particles. Conversely, coarser mixes have less total aggregate surface area and typically demand less asphalt.

The relationship between aggregate surface area and optimum asphalt content is most pronounced when mineral filler containing high percentages of particles passing the No. 200 (.075 millimeter) sieve is part of the mix. Variations in the amount and the size of mineral filler at a constant asphalt content will cause changes in mix properties, resulting in mixes with too little or too much asphalt. Small increases in filler content will use up additional asphalt and produce a dry, unstable mix. Small decreases have the opposite effect. Too little filler results in a mixture that is too rich.

An abundance of very fine (less than 20 microns) particles in the filler will act as an asphalt extender, resulting in a mix that has too much asphalt.

If a mix contains too little or too much mineral filler, arbitrary adjustments are likely to have adverse effects. Instead, proper sampling and testing should be done to determine the cause of the variations and, if necessary, to establish a new mix design.

Aggregate absorption is critical in determining optimum asphalt content because enough asphalt must be added to the mix to allow for absorption while still coating the particles with an adequate film. The absorption for established aggregate sources is generally known but requires careful testing when using new aggregate sources.

Two terms in HMA technology are used to express the methods of asphalt content: total asphalt content and effective asphalt content. Total asphalt content is the amount of asphalt that must be added to the mixture to produce the desired mix qualities. Effective asphalt content is based on the volume of asphalt not absorbed by the aggregate—of asphalt that effectively forms a bonding film on the aggregate surfaces. Effective asphalt content is calculated by subtracting the amount of absorbed asphalt, multiplied by the percentage of aggregate, from the total asphalt content.

4.1.4 Objectives and Applications of HMA Testing

The testing of an HMA mixture has three important applications with respect to the overall production of a quality HMA roadway. These are preliminary design testing, job-mix control testing, and routine construction control.

Preliminary Design Testing

The principal purpose of preliminary design testing is to determine if prospective sources can provide aggregate of satisfactory quality to produce a paving mix satisfying both the gradation and mix design requirements of the specifications. Asphalt also should meet specification requirements and be compatible with the available aggregates. Initial trial mixes should be made with an aggregate gradation approaching the median of the grading limits. Where several sources of aggregate are being considered, trial blends can determine the most economical combination to use. The results of the preliminary mix design serve as a basis for making a preliminary cost estimate. These results also indicate whether the design requirements can be obtained within the framework of the specifications.

Source acceptance testing is often performed after the contract has been awarded and the contractor has indicated the proposed sources of all aggregate materials and asphalt. This assures the selection of proper materials and permits the contractor to begin stockpiling these materials at the job site.

Job-Mix Formula

During the development of the mix design, testing is conducted to determine the appropriate proportions of the various aggregate fractions to result in an HMA mixture that will meet the specifications for the project. The result of this testing is the development of a job-mix formula for the project. The job-mix formula identifies the gradation of the combined aggregate materials and the selected asphalt content. In producing asphalt-paving mixture in a plant, certain variations in the mix are inevitable. A job-mix formula often contains tolerances that allow reasonable variations in the gradation and asphalt content.

Job-Mix Control Testing

Job-mix control testing is performed at the start of plant production and in conjunction with calibrating the central mixing plant for the job-mix formula. This testing is done on HMA mixtures produced through the HMA plant being used on the project and with materials at the job site. This is required because the process of producing the HMA mix through the plant may change the properties of the aggregate. This also provides an opportunity to ensure that the plant is properly calibrated. Such testing serves to determine whether the paving mixture produced under the job-mix formula meets all specification requirements.

Routine Construction Control

Construction control testing is performed as a routine and periodic item of inspection during paving construction. Random samples of the hot paving mixture are obtained and tested for design properties. The results of these tests are compared with the job-mix control tests and the overall specification requirements. Where irregularities occur and the limits of the job-mix formula are exceeded, appropriate corrections are required at the plant. Although this type of testing is routine in nature, it requires close attention. Its importance is emphasized because the results serve as a basis for the final acceptance or rejection of the paving construction.

4.2 Volumetric Calculations

4.2.1 Introduction

The following analytical procedures apply to paving mixtures compacted in a laboratory and to undisturbed samples cut from compacted pavement.

By analyzing a compacted paving mixture for air-voids and voids in the mineral aggregate, some indication of its probable durability and in-service performance may be determined. Air-voids are the small air spaces between coated aggregate particles; VMA consists of the intergranular void spaces between the particles of aggregate that hold asphalt in a compacted mixture. They are expressed as a percentage of the total volume of the compacted mixture.

It is possible to develop many equations for use in density-voids analyses. When the basic concepts are understood, the development of new equations to fit a particular set of measurements is a relatively simple procedure.

Information regarding the ingredients of an asphalt paving mixture must be known before any weight-volume calculations are made. This information includes the specific gravity of the asphalt, the bulk specific gravity for each aggregate, and the proportions of each ingredient in the mix.

In addition, a set of weight measurements must be made on an uncompacted sample of mix for determining the amount of asphalt absorbed by the aggregate. With this information, it is possible to make an analysis of density-voids relationships of compacted asphalt paving specimens whose weight and bulk volume are known.

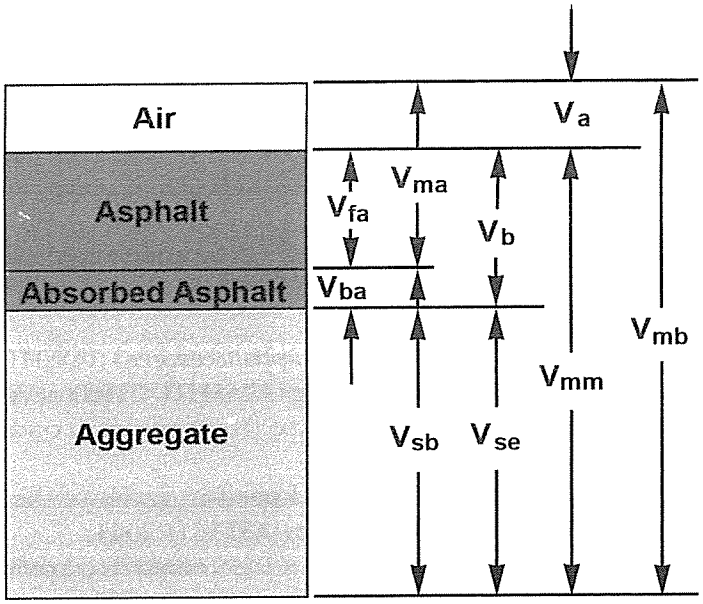
To aid in understanding the fundamentals involved, a block diagram of asphalt paving mixture is shown in Figure 4.3. The properties are shown in terms of volume, but they may be expressed in weights also, using the same subscripts. Figure 4.3 should be referred to as equations are developed.

4.2.2 Definitions

Air-voids (V_a) is the total volume of the small pockets of air between the coated aggregate particles throughout a compacted paving mixture, expressed as a percentage of the total volume of the compacted paving mixture.

Voids in the mineral aggregate (VMA) is the volume of intergranular void space between the uncoated aggregate particles of a compacted paving mixture that includes the air-voids and the effective asphalt content. VMA is expressed as a percentage of the total volume of the compacted paving mixture.

Voids filled with asphalt (VFA) is the percentage of the VMA that is occupied by the effective asphalt. It is expressed as the ratio of (VMA – V_a) to VMA.



V_{ma} = Volume of voids in mineral aggregate
V_{mb} = Bulk volume of compacted mix
V_{mm} = Voidless volume of paving mix
V_{fa} = Volume of voids filled with asphalt
V_a = Volume of air voids
V_b = Volume of asphalt
V_{ba} = Volume of absorbed asphalt
V_{sb} = Volume of mineral aggregate (by bulk specific gravity)
V_{se} = Volume of mineral aggregate (by effective specific gravity)

Figure 4.3 Hot Mix Asphalt Component Diagram

Asphalt content (P_b) is the total asphalt content of a paving mixture.

Absorbed asphalt content (P_{ba}) is the portion of asphalt absorbed into the aggregate particles.

Effective asphalt content (P_{be}) is the total asphalt content of a paving mixture minus the portion of asphalt absorbed into the aggregate particles.

4.2.3 Analyzing a Compacted Pavement Mixture

Data for all illustrative calculations in this section are based on the information given in Table 4.1. All the density-voids calculations involve

measurements of the materials at 77°F (25°C), including specific-gravity tests. In some cases, submerged weights are necessary. Even though the density of water is 62.416 pounds per cubic feet (0.997 grams per cubic centimeter) at 77°F (25°C), it is assumed to be 62.4 pounds per cubic feet (1 gram per cubic centimeter). The error induced by this assumption is considered insignificant.

The following is an outline of steps to use in the analysis of paving mixtures:

1. Measure the bulk specific gravities of the coarse aggregate (AASHTO T 85 or ASTM C 127) and of the fine aggregate (AASHTO T 84 or ASTM C 128).
2. Measure the specific gravity of the asphalt cement (AASHTO T 228 or ASTM D 70) and of the mineral filler (AASHTO T 100 or ASTM D 854).
3. Calculate the bulk specific gravity of the aggregate combination in the paving mixture.
4. Measure the maximum theoretical specific gravity of the loose paving mixture (AASHTO T 209 or ASTM D 2041).
5. Measure the bulk specific gravity of the compacted paving mixture (AASHTO T 166 or ASTM D 1188 or ASTM D 2726).
6. Calculate the effective specific gravity of the aggregate.
7. Calculate the asphalt absorption of the aggregate.
8. Calculate the effective asphalt content of the paving mixture.
9. Calculate the percentage of voids in the mineral aggregate in the compacted paving mixture.
10. Calculate the percentage of air-voids in the compacted paving mixture.
11. Calculate the percentage of voids filled with asphalt of the compacted paving mixture.

Table 4.1 illustrates the basic data for a sample of paving mixture. These design data are used in the sample calculation employed throughout the remainder of the chapter.

Table 4.1
Basic Data for Sample of Paving Mixture

Material	Specific Gravity	Bulk	Mix Composition		Percent by Weight of Total Mix	Percent by Weight of Total Aggregate
			AASHTO Method	ASTM Method		
Asphalt Cement	1.010 (G_b)	–	T 228	D 70	6.96 (P_b)	7.48 (P_b)
Coarse Aggregate		2.606 (G_1)	T 85	C 127	51.45 (P_1)	55.30 (P_1)
Fine Aggregate		2.711(G_2)	T 84	C 128	34.24 (P_2)	36.80 (P_2)
Mineral Filler	2.697 (G_3)	–	T 100	D 854	7.35 (P_3)	7.90 (P_3)

Bulk specific gravity of compacted paving mixture sample, G_{mb} (AASHTO T 209 or ASTM D 2041) 2.344
Maximum specific gravity of paving mixture sample, G_{mm} (AASHTO T 166, ASTM D 1188 or ASTM D 2727) 2.438

Apparent Specific Gravity of Aggregate

Apparent specific gravity (G_{sa}) (Figure 4.4) is the ratio of the mass in air of a unit volume of an impermeable material at a stated temperature to the mass in air of equal density of an equal volume of gas-free distilled water at a stated temperature. In other words, the aggregate apparent specific gravity does not include the volume of the water permeable voids in the aggregate.

Bulk Specific Gravity of Aggregate

Bulk specific gravity (G_{sb}) (Figure 4.5) is the ratio of the mass in air of a unit volume of a permeable material (including both permeable and impermeable voids normal to the material) at a stated temperature to the mass in air of equal density of an equal volume of gas-free distilled water at a stated temperature. In other words, the aggregate bulk specific gravity *includes* the volume of the water permeable voids in the aggregate (often termed the *saturated surface* dry or *SSD* volume of the aggregate). The test procedure for determining the bulk specific gravity of the fine and coarse aggregate fractions is discussed in Chapter 3, "Mineral Aggregates and Mixture Additives."

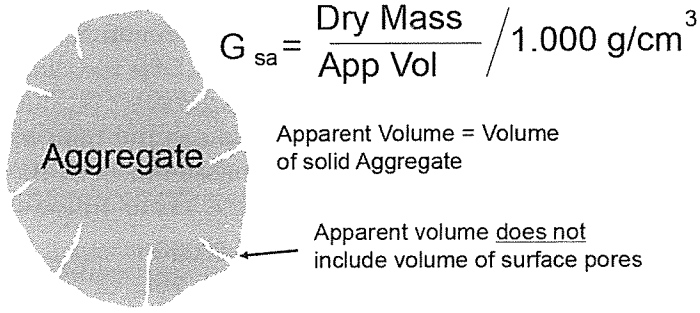


Figure 4.4 Apparent Specific Gravity of Aggregate

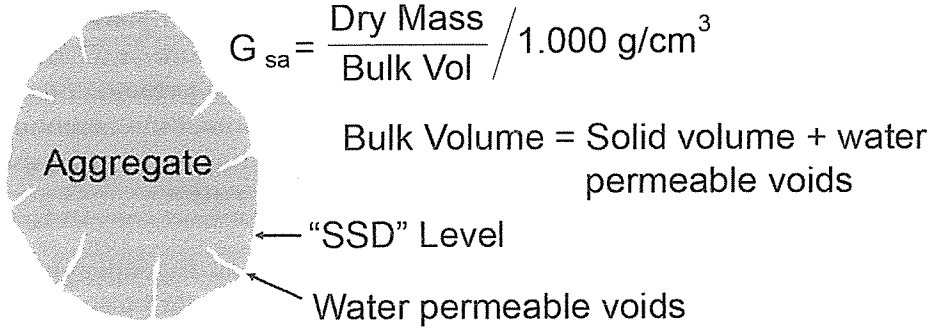


Figure 4.5 Bulk Specific Gravity of Aggregate

Combining Aggregates of Different Specific Gravities

When the total aggregate consists of separate fractions of coarse aggregate, fine aggregate, and mineral filler—all having different specific gravities—the bulk specific gravity for the aggregate combination is calculated as follows:

$$G_{sb} = \frac{P_1 + P_2 + \cdots + P_n}{\frac{P_1}{G_1} + \frac{P_2}{G_2} + \cdots + \frac{P_n}{G_n}}$$

where

G_{sb} = bulk specific gravity for the total aggregate
P_1, P_2, P_n = percentages by weight of aggregates 1, 2, n; and
G_1, G_2, G_n = bulk specific gravities of aggregates 1, 2, n

The bulk specific gravity of mineral filler is difficult to determine accurately at the present time. However, if the apparent specific gravity of the filler is used instead, the error is usually negligible.

The calculation using the data in Table 4.1 is as follows:

$$G_{sb} = \frac{51.450 + 34.240 + 7.350}{\frac{51.450}{2.606} + \frac{34.240}{2.711} + \frac{7.350}{2.697}} = \frac{93.040}{19.743+12.630+2.725} = 2.651$$

If the apparent specific gravity of the coarse aggregate is 2.759 and the apparent specific gravity of the fine aggregate is 2.905 for the sample of paving mix in Table 4.1, the apparent specific gravity (G_{sa}) of the total aggregate can be calculated by the same formula as the bulk by using the apparent specific gravity of each aggregate constituent. For this example, then, the calculated apparent specific gravity (G_{sa}) is as follows:

$$G_{sa} = \frac{51.450 + 34.240 + 7.350}{\frac{51.450}{2.759} + \frac{34.240}{2.905} + \frac{7.350}{2.697}} = \frac{93.040}{33.160} = 2.806$$

Maximum Specific Gravity of the HMA Mixture, AASHTO T 209 or ASTM D 2041

Maximum theoretical specific gravity (G_{mm}) is the ratio of the mass in air of a unit volume of the asphalt and aggregate in the mixture at a stated temperature to the mass in air of equal density of an equal volume of gas-free distilled water at a stated temperature. In other words, the maximum theoretical specific gravity (G_{mm}) is the mass of the asphalt and aggregate mixture divided by the volume of the asphalt coated particles.

Test Procedure

There are three basic steps in determining the maximum specific gravity (see Figure 4.6). The loose mix is warmed and separated into loose, individually coated aggregates. A predetermined mass of the dry loose mix is placed in a metal bowl or calibrated pycnometer and covered with water. A vacuum lid is fitted and secured to the bowl or pycnometer and placed on a vibratory shaker table. A vacuum pump is started and the manometer reading used to determine the proper vacuum adjustment. Once the proper

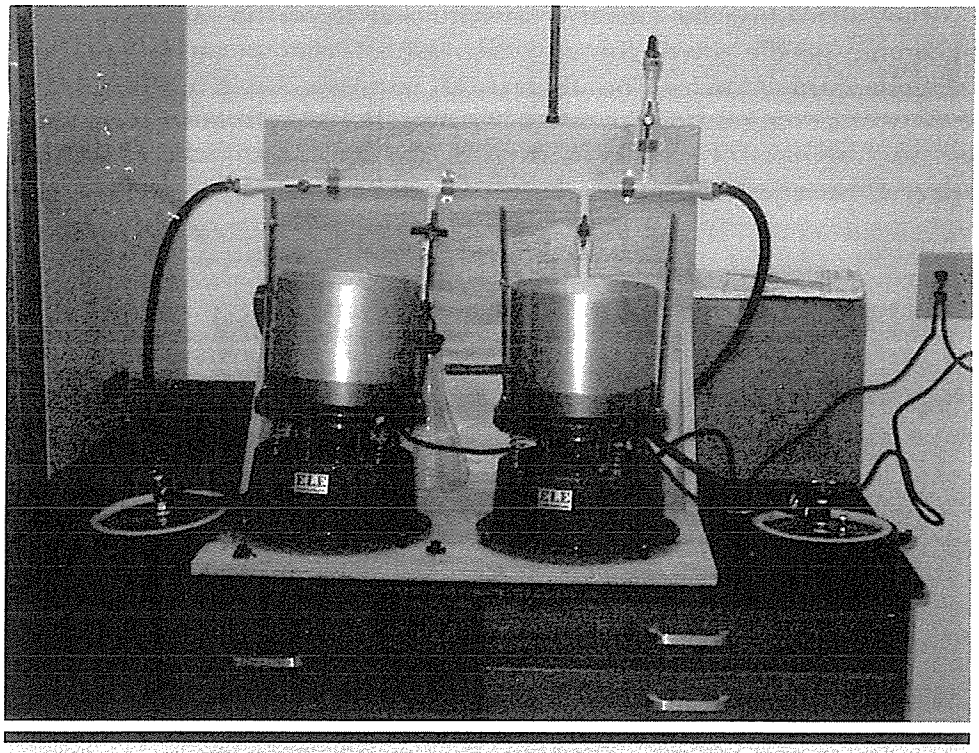


Figure 4.6 Equipment for Maximum Specific Gravity

partial vacuum is obtained, the shaker table is started. This provides gentle agitation to help in the removal of any air between particles. The agitation ensures that the air in the mixture is as close as possible to zero. Figure 4.6 depicts one of several different methods to determine G_{mm}. The theoretical maximum specific gravity is calculated using the equation shown for the specific procedure utilized. G_{mm} is the mass of the coated aggregate divided by the volume of coated aggregate. Air-voids are calculated from the bulk and maximum specific gravities. The ratio of these two specific gravities is actually the volume percentage of solids (in decimal form).

Maximum Specific Gravities of Mixtures with Different Asphalt Contents

In designing a paving mixture with a given aggregate, the maximum specific gravities (G_{mm}) at different asphalt contents are needed to calculate the percentage of air-voids for each asphalt content. While individual maximum specific gravities can be determined for each asphalt content, the precision of the test is best when the specimen is at, or over the optimum asphalt content. The Asphalt Institute recommends measuring the maximum specific gravity in duplicate or triplicate on individually batched specimens at one trial asphalt content. The end result of the maximum specific gravity test is to determine an accurate G_{se} (Effective Specific Gravity of Aggregate) as described in the following section. Many agencies have different procedures to determine G_{se}. The designer is encouraged to be familiar with methods in their respective localities.

After calculating the effective specific gravity of the aggregate, the maximum specific gravity for any other asphalt content can be obtained using the formula below. For all practical purposes, the effective specific gravity of the aggregate is constant because the asphalt absorption does not vary appreciably with variations in asphalt content in the working range of the mix design formula.

$$G_{mm} = \frac{P_{mm}}{\frac{P_s}{G_{se}} + \frac{P_b}{G_b}}$$

where

G_{mm} = maximum specific gravity of paving mixture (no air-voids)
P_{mm} = total loose mixture, percent by total weight of mixture = 100 percent
P_s = aggregate, percent by total weight of mixture
P_b = asphalt, percent by total weight of mixture
G_{se} = effective specific gravity of aggregate
G_b = specific gravity of asphalt

Calculations using the specific gravity data from Table 4.1, the effective specific gravity (G_{se}), determined on the previous page and an asphalt content (P_b) of 7.46 percent:

$$G_{mm} = \frac{100}{\frac{92.540}{2.726} + \frac{7.460}{1.010}} = \frac{100}{41.33} = 2.419$$

Effective Specific Gravity of Aggregate

Effective specific gravity (G_{se}) is the ratio of the mass in air of a unit volume of a permeable material (excluding voids permeable to asphalt) at a stated temperature (see Figure 4.7) to the mass in air of equal density of an equal volume of gas-free distilled water at a stated temperature. In other words, the effective specific gravity includes the volume of the water-permeable voids in the aggregate that cannot be reached by the asphalt.

The effective specific gravity (G_{se}) of the aggregate includes all void spaces in the aggregate particles except those that absorb asphalt. It is determined as follows:

$$G_{se} = \frac{P_{mm} - P_b}{\frac{P_{mm}}{G_{mm}} - \frac{P_b}{G_b}}$$

where

G_{se} = effective specific gravity of aggregate
P_{mm} = total loose mixture, percent by total weight of mixture = 100 percent

$$G_{se} = \frac{\text{Dry Mass}}{\text{Eff Vol}} \Big/ 1.000\ \text{g/cm}^3$$

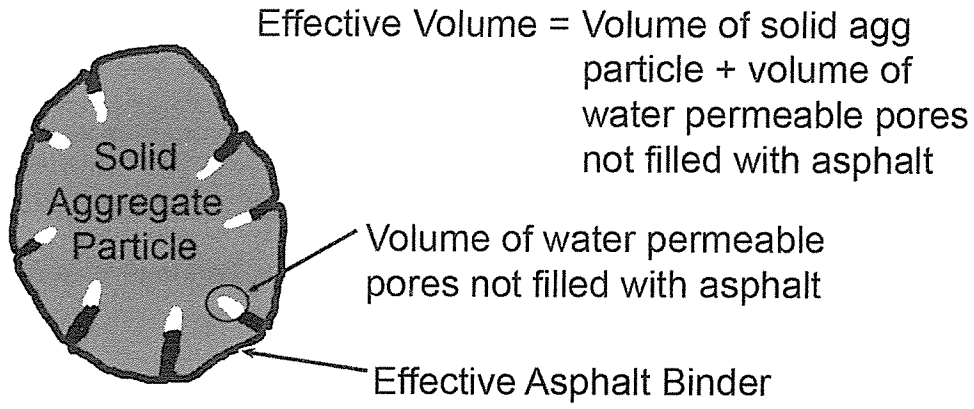


Figure 4.7 Effective Specific Gravity of Aggregate

P_b = asphalt, percent by total weight of mixture
G_{mm} = maximum specific gravity of paving mixture (no air-voids), AASHTO T 209 or ASTM D 2041
G_b = specific gravity of asphalt

Calculation using the data in Table 4.1 is as follows:

$$G_{se} = \frac{100 - 6.960}{\frac{100}{2.438} - \frac{6.960}{1.010}} = \frac{93.040}{34.126} = 2.726$$

The volume of asphalt binder absorbed by an aggregate is always less than the volume of water absorbed. Consequently, the value for the effective specific gravity of an aggregate should be between its bulk and apparent specific gravities. When the effective specific gravity falls outside these limits, its value must be assumed to be incorrect. The maximum specific gravity of the total mix, the aggregate gradation and total asphalt content should then be checked for possible errors. The G_{se} is needed to calculate asphalt absorption and to back-calculate the G_{mm} at other asphalt percentages.

In the example, the three specific gravities are as follows:

Bulk Specific Gravity (G_{sb})	2.651
Effective Specific Gravity (G_{se})	2.726
Apparent Specific Gravity (G_{sa})	2.806

Bulk Specific Gravity of the HMA Mixture

The bulk specific gravity of a compacted sample is the mass of aggregate and binder divided by the bulk volume (i.e., volume of aggregate, binder, *and* air). Two methods are used to determine the bulk specific gravity of a mix. The first is the AASHTO T 166 (ASTM D 2726) procedure and the second is the CoreLok® procedure.

Bulk Specific Gravity, AASHTO T 166 or ASTM D 2726 Procedure

The steps for determining the mixture bulk specific gravity are given in this test procedure. After mixing, aging, and compacting the mixture, the mass of the sample is determined in air (dry), while submerged in water, and then in air again after drying the surface (saturated surface dry). The mass of the oven-dry specimen is being determined in Figure 4.8. The next step is to place the specimen in the water bath directly below the scale (not shown) and determine its mass under water. The last step is to determine the mass of the saturated surface dry specimen in air. The saturated surface-dry (SSD) mass is obtained by quickly blotting the sample so that the surface is still wet but not shiny (see Figure 4.9). The bulk specific gravity is the mass of the sample divided by the mass (volume) of water it displaces.

Bulk Specific Gravity, CoreLok® Procedure

The CoreLok® test is conducted by the water-displacement method using saturated surface-dry (SSD) specimens. The SSD method has been proven to work for dense-graded HMA mixtures. A problem may occur with coarse-graded mixtures. These mixes tend to have larger internal air-voids than the conventional mixes, even though the volume of air-voids is the same. Mixes with coarser (or gap-graded) aggregates will have a higher percentage of large aggregate particles. At some overall air volume, which is mix-specific, the large internal air-voids of the coarse aggregate can become interconnected. During the bulk specific gravity testing with the SSD method, water can quickly infiltrate the sample. After removing



Figure 4.8 Determination of Dry Mass of Specimen

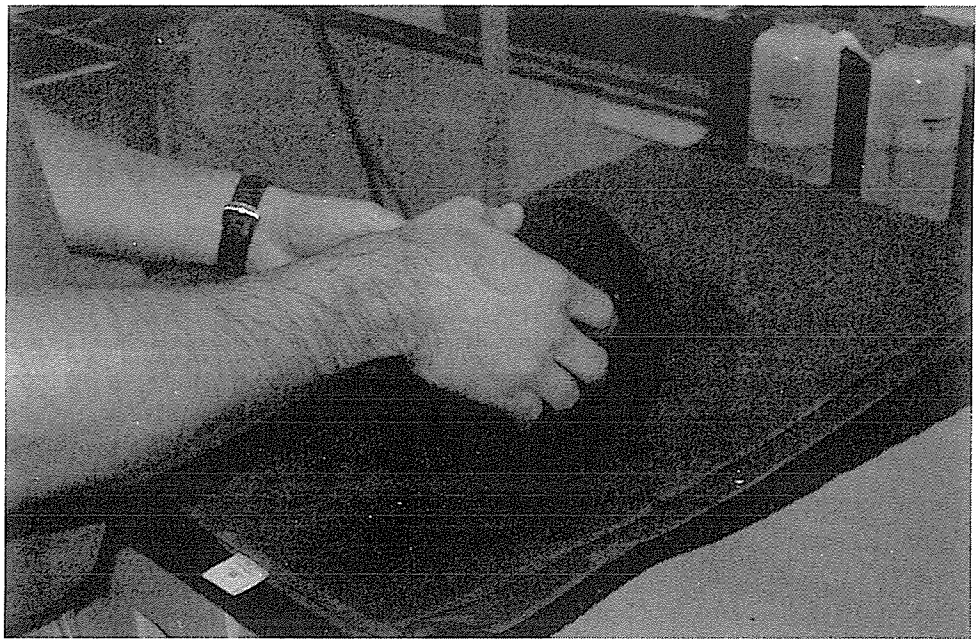


Figure 4.9 Determination of SSD Mass of Specimen

the sample from the water bath to obtain the SSD condition, the water may also drain from the sample quickly. This draining of the water from the sample can cause errors with the water-displacement method.

To overcome this problem, recent studies have centered on the use of the CoreLok® vacuum-sealing device (see Figure 4.10). This device has been shown to be more accurate than conventional methods such as water displacement, Parafilm, and dimensional analysis. The CoreLok® device uses an automatic vacuum chamber with a puncture-resistant bag (Figure 4.11), which tightly conforms to the sides of the sample and prevents water from infiltrating the sample.

The water-displacement procedure (AASHTO T 166 or ASTM D 2726) can overestimate the density of a hot mix asphalt sample. This overestimation is very small for fine-graded mixes, increases for coarse-graded mixes, and increases again for gap-graded mixes such as SMA.

Asphalt Absorption

Absorption is expressed as a percentage by weight of aggregate rather than as a percentage by total weight of mixture. Asphalt absorption (P_{ba}) is determined as follows:

$$P_{ba} = 100\left(\frac{G_{se} - G_{sb}}{G_{sb}G_{se}}\right)G_b$$

where

P_{ba} = absorbed asphalt, percent by weight of aggregate
G_{se} = effective specific gravity of aggregate



Figure 4.10 CoreLok Device



Figure 4.11 HMA Sample in Sealed Bag

G_{sb} = bulk specific gravity of aggregate
G_b = specific gravity of asphalt

The calculation using specific gravity data from Table 4.1 is as follows:

$$P_{ba} = 100\left(\frac{2.726 - 2.651}{2.651 \times 2.726}\right)1.010 = 100\left(\frac{0.075}{7.227}\right)1.010 = 1.05$$

Effective Asphalt Content of a Paving Mixture

The effective asphalt content (P_{be}) of a paving mixture is the total asphalt content minus the quantity of asphalt lost by absorption into the aggregate particles. It is the portion of the total asphalt content that remains as a coating on the outside of the aggregate particles, and it is the asphalt content on which service performance of an asphalt paving mixture depends. The formula is as follows:

$$P_{be} = P_b - \left(\frac{P_{ba}}{100}\right)P_s$$

where

P_{be} = effective asphalt content, percent by total weight of mixture
P_b = asphalt, percent by total weight of mixture
P_{ba} = absorbed asphalt, percent by weight of aggregate
P_s = aggregate, percent by total weight of mixture

The following is an example using data calculated in the previous examples:

$$P_{be} = 6.96 - \left(\frac{1.05}{100}\right) 93.04 = 6.96 - 0.98 = 5.98$$

Percent VMA in Compacted Paving Mixture

The voids in the mineral aggregate (VMA) are defined as the intergranular void space between the aggregate particles in a compacted paving mixture that includes the air-voids and the effective asphalt content, expressed as a percent of the total mix volume. The VMA is calculated on the basis of the bulk specific gravity of the aggregate and is expressed as a percentage of the bulk volume of the compacted paving mixture. Therefore, the VMA can be calculated by subtracting the volume of the aggregate determined by its bulk specific gravity from the bulk volume of the compacted paving mixture. The method of calculation is illustrated here.

If mix composition is determined as percent by weight of total mixture, then

$$VMA = 100 - \left(\frac{G_{mb}P_s}{G_{sb}}\right)$$

where

VMA = voids in mineral aggregate (percent of bulk volume)
G_{sb} = bulk specific gravity of aggregate
G_{mb} = bulk specific gravity of compacted mixture (AASHTO T 166 or ASTM D 2726)
P_S = aggregate, percent by total weight of mixture

The calculation using data from Table 4.1 is as follows:

$$VMA = 100 - \frac{2.344 \times 93.04}{2.651} = 100 - 82.27 = 17.73$$

Or if mix composition is determined as percent by weight of aggregate, then

$$VMA = 100 - \left(\frac{G_{mb}}{G_{sb}} \times \frac{100}{100+P_b}\right) 100$$

where

P_b = asphalt, percent by weight of aggregate

The calculation using data from Table 4.1 is as follows:

$$VMA = 100 - \left(\frac{2.344}{2.651}\right)\left(\frac{100}{100+7.48}\right)(100) = 100 - 82.27 = 17.73$$

Calculation of Air-Voids in Compacted Mixture

The air-voids (P_a) in a compacted paving mixture consist of the small air spaces *between* the coated aggregate particles. The percentage of air-voids in a compacted mixture can be determined using the following equation:

$$P_a = 100\left(\frac{G_{mm} - G_{mb}}{G_{mm}}\right)$$

where

P_a = air-voids in compacted mixture, percent of total volume
G_{mm} = maximum specific gravity of paving mixture (as determined in previous discussion, or as determined directly for a paving mixture by AASHTO T 209 or ASTM D 2041)
G_{mb} = bulk specific gravity of compacted mixture

The calculation using data from Table 4.1 is as follows:

$$P_a = 100\left(\frac{2.438-2.344}{2.438}\right) = \frac{9.400}{2.438} = 3.86$$

Percent VFA in Compacted Paving Mixture

The voids filled with asphalt (VFA) is the percentage of the intergranular void space between the aggregate particles (VMA) that are filled with asphalt. VFA, not including the absorbed asphalt, is determined using this formula:

$$VFA = \frac{100(VMA - P_a)}{VMA}$$

where

VFA = voids filled with asphalt, percent of VMA
VMA = voids in the mineral aggregate, percent of bulk volume
P_a = air-voids in compacted mixture, percent

The calculation using data from Table 4.1 is as follows:

$$VFA = \frac{100(17.73-3.86)}{17.73} = 78.2\%$$

4.3 Superpave HMA Mix Design System

4.3.1 Introduction

In 1987, a five-year FHWA study to improve the procedures for the performance of HMA pavements was conducted. The result was the Superpave (SUPerior PERforming Asphalt PAVEments) system. This system was developed to give pavement engineers and contractors the tools they needed to improve the performance of HMA pavements subjected to extremes of traffic loading and temperature. The Superpave system as implemented consisted of two interrelated elements: an asphalt binder specification (discussed in Chapter 2, "Petroleum Asphalts") and a mix design and analysis system. Since the implementation of Superpave in the mid-1990s, research has continued on the development of performance models. The status of that research is summarized in Section 4.8 in this chapter. The following summarizes the Superpave mix design system:

The Superpave mix design procedure consists of:

1. Selecting asphalt and aggregate materials that meet the design criteria
2. Developing several aggregate trial blends that will meet the Superpave gradation requirements
3. Blending the asphalt binder with the aggregate trial blends and short-term oven aging of the mix
4. Compacting the specimens
5. Analyzing the volumetric properties of the mixture (using the procedures described in Section 4.2)
6. Selecting the best aggregate blend as the design aggregate structure based on the desired performance criteria for the mixture

A detailed discussion is provided in the Asphalt Institute's SP-2, *Superpave Mix Design*. The detailed procedures for conducting the calculations associated with Superpave mix design are included in SP-2 and the *Mix Design Program*, SW-2. This program is written in strict conformance with the SP-2 manual. It is an Excel-based program that allows users to modify the program components to their needs.

4.3.2 Materials Selection

Asphalt Binder

The asphalt binder to be used in the mix is chosen based on environmental and traffic loadings using the procedures described in Chapter 2, "Petroleum Asphalts."

Mineral Aggregate

During the development of Superpave, pavement experts agreed that certain aggregate characteristics were critical to a well-performing HMA. These characteristics were broken into two categories: consensus properties and source properties. A detailed discussion of these properties and test procedures is included in Chapter 3, "Mineral Aggregates and Mixture Additives.

Consensus Aggregate Properties

The consensus aggregate properties used in the Superpave mix design procedure are the following: coarse aggregate angularity, fine aggregate angularity, flat and elongated particles, and clay content. The criteria for these properties are based on the traffic level and position within the pavement structure. The materials near the pavement surface are subjected to higher stress levels and require materials of higher quality. The criteria should generally be applied to a proposed aggregate blend rather than individual components. The desired consensus properties are detailed in Table 4.2.

Table 4.2
Superpave Aggregate Consensus Requirements

Design ESALs[1] (million)	Coarse Aggregate Angularity (Percent), minimum[2]		Uncompacted Void Content of Fine Aggregate (Percent), minimum		Sand Equivalent (Percent), minimum	Flat and Elongated[3] (Percent), maximum
	≤ 100 mm	> 100 mm	≤ 100 mm	> 100 mm		
< 0.3	55/–	–/–	–	–	40	–
0.3 to < 3	75/–	50/–	40	40	40	10
3 to < 10	85/80	60/–	45	40	45	10
10 to < 30	95/90	80/75	45	40	45	10
≥ 30	100/100	100/100	45	45	50	10

[1] Design ESALs are the anticipated project traffic level expected on the design lane over a 20-year period. Regardless of the actual design life of the roadway, determine the design ESALs for 20 years and choose the appropriate N_{design} levels.

[2] 85/80 denotes that 85 percent of the coarse aggregate has one fractured face and 80 percent has two or more fractured faces.

[3] Criterion based upon a 5:1 maximum-to-minimum ratio.
(If less than 25 percent of a layer is within 100 millimeters of the surface, the layer may be considered to be below 100 millimeters for the mixture-design purposes.)

Source Aggregate Properties

The source properties used in the Superpave mix design procedure are the following: toughness or Los Angeles Abrasion, sodium or magnesium sulfate soundness and deleterious materials. These properties are source specific, and the requirements vary throughout the United States because of the variety of aggregate types available in the various geographical areas. The requirement for a maximum loss value from the Los Angeles Abrasion test will typically range from 35 percent to 45 percent.

The requirement for maximum loss value for the soundness test will typically range from 10 percent to 20 percent for five cycles. A wide range of criteria for maximum allowable percentage of deleterious particles exists. Values range from as low as 0.2 percent to as high as 10 percent.

Gradation

Superpave utilizes the 0.45 power gradation chart to analyze the aggregate gradation and to define the maximum density line. The maximum density line represents a gradation where the aggregate particles fit together in their densest possible arrangement. The development and use of this chart is discussed in Chapter 3. Superpave uses the maximum and nominal maximum size of the aggregates to categorize mixtures and define gradation requirements. These are defined as follows:

- Maximum size: One sieve size larger than the nominal maximum size
- Nominal maximum size: One sieve size larger than the first sieve to retain more than 10 percent

4.3.3 Design Aggregate Structure

The term used in Superpave to describe the distribution of aggregate particle sizes is *design aggregate structure.* A design aggregate structure that lies between the control points meets the Superpave gradation requirements. Superpave defines six mixture gradations based on their nominal maximum aggregate size (see Table 4.3). The SP-2 manual provides the numerical gradation limits for the five Superpave mixtures.

Table 4.3
Superpave Mixture Gradations

Superpave Designation, mm	Nominal Maximum Size, mm	Maximum Size, mm
37.5	37.5	50.0
25.0	25.0	37.5
19.0	19.0	25.0
12.5	12.5	19.0
9.5	9.5	12.5
4.75	4.75	9.5

Select Asphalt and Aggregate Materials

The asphalt binder is selected based on the environmental and traffic conditions as discussed in Chapter 2, "Petroleum Asphalt."

The aggregate materials are chosen based on the blended materials meeting the consensus and source requirements as specified by the agency. To select a specific design aggregate structure, the designer establishes trial blends by mathematically combining the gradation of the individual materials into a single gradation. The blended gradation is compared to the specification

requirements for the appropriate sieves. Many designers evaluate multiple blends. The number of blends evaluated varies depending on the experience of the designer with the aggregate source being used on the project. If the source is new, it will require more blends than a source the designer has been using for many years. A detailed example of the process for selecting a design aggregate structure is included in the SP-2 manual.

Select Initial Asphalt Binder Content

When the aggregate blends have been chosen, the next step is to evaluate the trial blends by compacting the specimens and determining the volumetric properties for each trial blend. For each blend, a minimum of two specimens is compacted using the Superpave Gyratory Compactor (SGC). The trial asphalt binder content can be estimated based on the designer's experience with similar materials. If the designer does not have experience with the particular aggregate source, the trial binder content can be determined for each trial blend by estimating the specific gravity of the blend and using a calculation procedure detailed in AASHTO R 35. SP-2 lists typical trial binder contents for aggregate blends having a combined bulk specific gravity of approximately 2.65.

Superpave Gyratory Compactor

The Superpave mix design procedure uses the Superpave Gyratory Compactor (see Figure 4.12) to compact the specimens. The basis for the SGC was the Texas gyratory compactor, which was modified to use the compaction principles of a French gyratory compactor. The Texas device accomplished the goal of achieving realistic specimen densification, and it was reasonably portable. Its 6-inch sample diameter (ultimately 150 millimeters on an SGC) could accommodate mixtures containing aggregate up to 50 millimeters maximum (37.5 nominal) size. The developers of the Superpave system modified the Texas device by lowering its angle and speed of gyration and adding real-time specimen-height recordation.

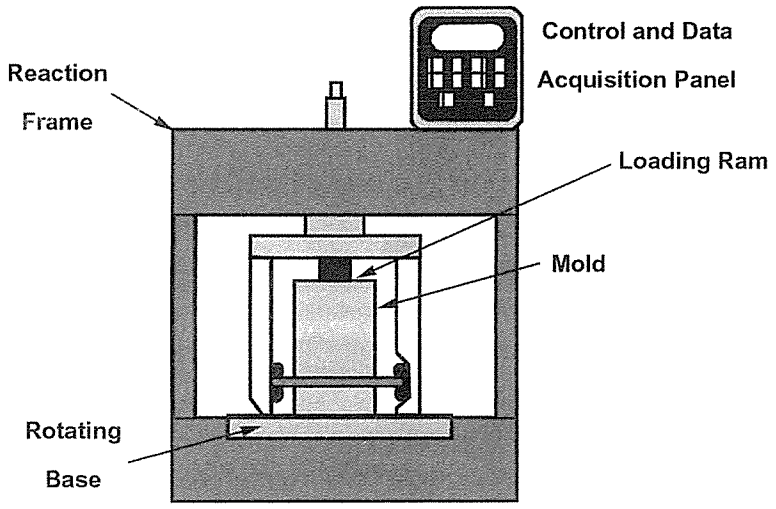


Figure 4.12 Superpave Gyratory Compactor

The SGC is a mechanical device comprising the following components:

- Reaction frame, rotating base, and motor
- Loading system, loading ram, and pressure gauge
- Height measuring and recordation system
- Mold and base plate
- A specimen extruding device

A loading mechanism presses against the reaction frame and applies a load to the loading ram to produce a 600-kiloPascals compaction pressure on the specimen. A pressure gauge measures the ram loading to maintain constant pressure during compaction. The SGC mold (see Figure 4.13) has an inside diameter of 150 millimeters, and a base plate in the bottom of the mold provides confinement during compaction. The SGC gyrates at a constant rate of 30 gyrations per minute with the mold positioned at an external compaction angle of 1.25 degrees. The internal angle is calibrated using procedures discussed below.

Specimen height measurement is an important function of the SGC. Using the mass of material placed in the mold, the diameter of the mold, and the specimen height, an estimate of specimen density can be made at any time throughout the compaction process. Specimen density is computed by dividing the mass by the volume of the specimen. The height is measured by recording the position of the ram throughout the test. Using these measurements, a specimen's compaction characteristic is developed. See Figure 4.14.

Figure 4.14 illustrates how the density of the asphalt mixture increases with increasing gyrations. As with other mix design procedures, asphalt mixtures are designed at a specific level of compaction effort. In the Superpave mix design system, this is a function of the design number of gyrations (N_{des}). N_{des} is used to vary the compactive effort of the design mixture, and it is a function of the traffic level. Traffic is represented by the design Equivalent Single Axle Loads (ESALs). The range of values for N_{des} is shown in Table 4.4.

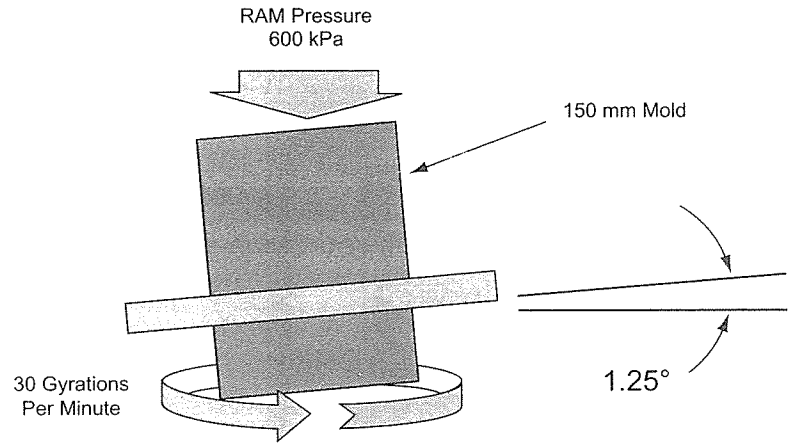


Figure 4.13 Superpave Gyratory Compactor Mold Configuration

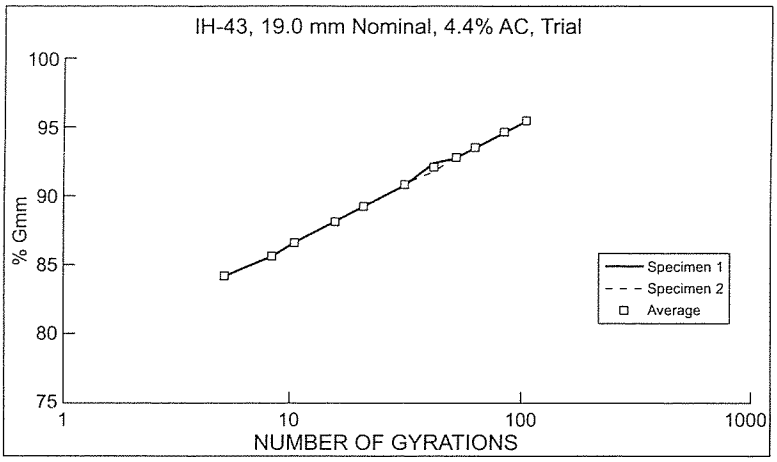


Figure 4.14 Mixture Compaction Characteristics from the SGC

Table 4.4
Superpave Gyratory Compactive Effort

Design ESALs[a] (million)	Compaction Parameters			Typical Roadway Applications
	$N_{initial}$	N_{design}	$N_{maximum}$	
< 0.3	6	50	75	Applications include roadways with very light traffic volumes, such as local roads, county roads, and city streets where truck traffic is prohibited or at a very minimal level. Traffic on these roadways would be considered local in nature, not regional, intrastate, or interstate. Special-purpose roadways serving recreational sites or areas may also be applicable to this level.
0.3 to 3	7	75	115	Applications include collector roads or access streets. Medium-trafficked city streets and the majority of country roadways may be applicable to this level.
3 to < 30	8	100	160	Applications include many two-lane, multilane, divided, and partially or completely controlled access highways. Among these are medium to heavily trafficked city streets, many state routes, U.S. highways, and some rural intersections.
> 30	9	125	205	Applications include the vast majority of the U.S. Interstate System, both rural and urban in nature. Special applications such as truck-weighing stations or truck-climbing lanes on two-lane roadways may also be applicable to this level.

[a] The significant project traffic level expected on the design lane over a 20-year period. Regardless of the design life of the roadway, determine the design ESALs for 20 years.